Forecasting Ottawa's Electricity Load for a Smarter Grid

Weather- and calendar-aware machine learning forecasts of hourly demand and baseload, with a net-load view of solar, wind and EV fleets

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The problem: what drives Ottawa's electricity load?

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Weather

Cold winters cause heating peaks; hot summers cause cooling peaks



Calendar

Workdays, weekends and statutory holidays change the daily load profile



Renewables

Solar and wind output changes from hour to hour with the weather



Electrification

EV fleets add large charging loads that can be scheduled

Research question

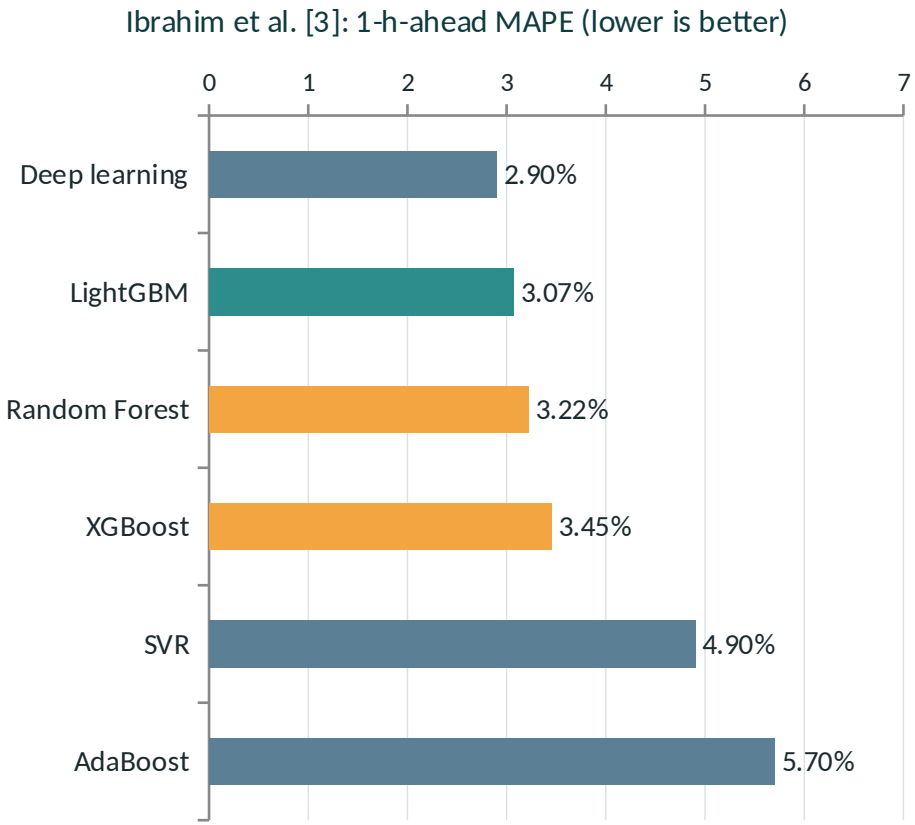
Can tree-ensemble models (Random Forest, XGBoost) forecast Ottawa's hourly electricity demand and next-day baseload more accurately than a Linear Regression baseline, using lagged demand, weather and calendar features?

Why it matters: an accurate forecast lets the grid schedule supply, storage and flexible loads (such as EV charging) ahead of demand, instead of reacting to it.

Baseload is the minimum level of demand that the grid must always serve. We define it as the lowest hourly load of each day.

State of the art: what recent studies found

Victor



Orange bars mark the two models we propose. Random Forest reached R^2 = 0.92 with 7.1 s of training, versus 20 min for deep learning.

Study	Data	Result
[1] Gökçe & Duman, IEEE IISEC 2022	Turkey, hourly, 2018–21 (35,064 h); temperature, holidays, 24-hour lag	RMSE: LR 2348, RF 2075, XGBoost 2039. RF had the lowest absolute and percentage errors
[2] Rawal & Ahmad, IEEE ICPC2T 2022	New South Wales, Australia, 2020; eight regression models	Non-parametric models outperformed parametric ones; Random Forest was best
[3] Ibrahim et al., Energies 2022	Panama, hourly, 2016–19 (33,600 h); temperature, humidity, lagged load	Deep learning R^2 = 0.93; RF and LightGBM R^2 = 0.92; 24-hour-ahead Bi-LSTM only 0.40
[4] Kong et al., IEEE Trans. Smart Grid 2019	Household smart-meter data	LSTM outperformed competing methods for individual households

How the literature shapes our design

Victor

Lagged load and temperature drive accuracy Load from the previous week and day, plus temperature, were the top predictors in [3]; [1] used a 24-hour lag.	**Lags t-1, t-24, t-168; temperature, humidity**
Calendar effects matter Both [1] and [3] used holiday and working-day indicators as inputs.	**Ontario statutory-holiday and workday flags**
Tree ensembles: near-best accuracy at low cost RF and XGBoost outperformed LR in [1]–[3], and came within 0.3 percentage points of deep learning's MAPE in [3].	**LR baseline vs Random Forest [9] vs XGBoost [10]**
Known weak spots Summer heat and COVID-19 reduced accuracy in [1]; 24-hour-ahead forecasts were much harder in [3].	**Report errors by season and horizon; flag the 2020 COVID period**

Gap we address: the four studies forecast a whole country, a state or a single home. None targets a city zone such as Ottawa, or examines how solar, wind and EV fleets change the net load left to serve.

Proposed solution: forecast, then balance

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1 · Real data

IESO hourly demand (Ottawa zone), ECCC and Open-Meteo weather, Ontario holidays

2 · Features

Lags t-1, t-24, t-168; temperature, HDD/CDD; hour, weekday, month, holiday flag

3 · ML models

Linear Regression (baseline), Random Forest, XGBoost

4 · Forecasts

Next-hour demand and next-day baseload (daily minimum)

5 · Scenario layer: live supply-demand view

Net load = forecast demand − solar farm − wind farm + EV fleet charging

Solar and wind output are estimated from weather data with physical models, and EV charging from workday and holiday schedules. The result shows, hour by hour, how much dispatchable supply is needed and when flexible EV charging can be shifted.

Data: real where possible, assumed where not

Victor

Component	Source	Resolution	Type
Ottawa-zone electricity demand (target)	IESO Hourly Zonal Demand Report, Ottawa zone [5]	Hourly, 2020–2025	**REAL**
Temperature, humidity	ECCC historical climate data, Ottawa station [6]	Hourly	**REAL**
Solar irradiance, 100 m wind speed	Open-Meteo historical weather API [7]	Hourly	**REAL**
Holidays, workdays	Ontario statutory holiday calendar	Daily	**REAL**
Solar farm output	pvlib PV model [8] driven by real irradiance	Hourly	**ASSUMED**
Wind farm output	Generic turbine power curve driven by real wind speed	Hourly	**ASSUMED**
Residential vs. industrial split, EV fleet load	Calendar-based profiles (next slide)	Hourly	**ASSUMED**

Data split: training 2020–2023, validation 2024, test 2025. The split is chronological with no shuffling, so no future data leaks into training.

Scenario assumptions: explicit and adjustable

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Solar farm

A 10 MW (AC) fixed-tilt array near Ottawa. Output is modelled with pvlib from real irradiance and temperature: zero at night, reduced under snow.

Wind farm

A 20 MW farm (e.g., six 3.3 MW turbines). Power curve: cut-in 3 m/s, rated 12 m/s, cut-out 25 m/s, applied to 100 m wind speed.

Residential

Follows temperature (heating and cooling degree-hours), with morning and evening peaks; higher on weekends and holidays.

Industrial / commercial

A flat daytime block on workdays, reduced on weekends and statutory holidays, and minimal overnight.

EV fleet

300 fleet vehicles using about 60 kWh each per workday, charged overnight (22:00–06:00); about 30% of that on holidays.

Calibration

Component shares are scaled so that they add up to the real IESO Ottawa-zone total. Every parameter can be varied for stress tests.

Methodology: models, baselines and evaluation

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Persistence / seasonal naive
The minimum bar for every model: demand(t+1) = demand(t), or the same hour last week

Linear Regression
An interpretable baseline that shows how much of the load is linear in temperature and calendar

Random Forest
An ensemble of bagged trees [9] that captures the nonlinear (U-shaped) heating and cooling response

XGBoost
Gradient-boosted trees [10]; achieved the lowest RMSE of the three models in [1]

Stretch goal: an LSTM sequence model [4], if time allows.

Evaluation on the unseen 2025 test year

MAE
MW

RMSE
MW

MAPE
%

R^2
goodness of fit

Critical analysis: errors by season, peak hours and holidays; feature importance; residual plots; and finally, net-load error once the solar, wind and EV scenarios are added.

Feasibility and timeline

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Why this scope is realistic

- The data is free, public and hourly: about 52,000 rows for 2020–2025, small enough to train in seconds to minutes on a laptop.
- We use standard, well-documented libraries: pandas, scikit-learn, XGBoost and pvlib.
- The core ML deliverable (steps 1–4) does not depend on the scenario layer, so the scenarios cannot put it at risk.
- Clear fallback: if the zonal data is incomplete, we will use IESO's Ontario-wide hourly demand with the same pipeline.
- Out of scope: real-time control, power-flow simulation of the grid, and private utility data.

Oct 13
Proposal (this talk)

Oct 14–31
Collect and clean data; build features; explore the data

Nov 1–15
Train the baselines, RF and XGBoost; tune on 2024 data

Nov 16–28
Build the scenario layer: solar, wind, EV and net load

Nov 29–Dec 8
Analyse the test year; prepare the final presentation

Dec 14
Submit the final report in IEEE format

Team contributions and use of AI

Kshitij
Problem definition and research question; smart-grid scenario layer (solar, wind, EV, net load); feasibility and timeline; slide design.

Victor
State-of-the-art review; data sources and preprocessing pipeline; model training and evaluation design.

Both members will share the coding and the writing of the final report.

AI disclosure

ChatGPT (OpenAI): creating summary tables and illustrative diagrams for our draft.

Claude (Anthropic), via Claude Code: restructuring the deck for the 5-minute limit, improving slide wording and layout, proofreading, checking citations, and drafting the scenario assumptions for our review.

Our own work: the choice of problem, papers, data sources and methods. We checked every number against the cited papers and take responsibility for all content.

References

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Questions?

Forecasting Ottawa's load from real data and balancing it against solar, wind and EV fleets